SQL Practice Task Sheet

Welcome, as your mentor, my goal is to help you master SQL from fundamentals to advanced concepts. The following task sheet is structured to ensure logical progression, practice, and exposure to real-world scenarios. You're encouraged to use any RDBMS (MySQL, PostgreSQL, SQL Server) as per your comfort.

1. SQL Basics

- Introduction to Databases and RDBMS Concepts
- Data Types and Table Schemas
- CRUD Operations:
  - Create tables and insert different types of data.
  - Retrieve data using `SELECT`.
  - Update and delete specific data records.
- Filtering & Sorting:
  - WHERE clause usages (text, numbers, dates).
  - Logical operators (`AND`, `OR`, `NOT`).
  - ORDER BY for sorting results.
- Practice: Set up a sample "Employees" and "Departments" table. Populate them and perform CRUD.

2. Aggregate Functions and Grouping

- Use of `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`.
- GROUP BY for aggregation.
- HAVING clause for filtered aggregations.
- Practice: Generate department-wise salary budgets, count employees per department, and filter groups.

3. Joins and Relationships

- Understanding relationships: One-to-One, One-to-Many, Many-to-Many.
- INNER JOIN vs LEFT/RIGHT/FULL OUTER JOIN.
- Practical scenarios for each type.

- CROSS JOIN and Self-Join.
- Practice: Join "Employees" with "Departments" and any other related tables.

4. Subqueries and Derived Tables

- Inline vs Correlated subqueries.
- Using subqueries in SELECT, FROM, and WHERE.
- Practice: Retrieve employees with above-average salaries, departments with more than five employees, etc.

5. Set Operations

- UNION, UNION ALL, INTERSECT, and EXCEPT/MINUS.
- Differences and use cases.
- Practice: Find employees not assigned to any department, merge lists from different tables.

6. Advanced SQL Functions

- String Functions: CONCAT, SUBSTRING, TRIM, UPPER/LOWER, REPLACE.
- Date and Time Functions: CURRENT_DATE, DATEADD, DATEDIFF, FORMAT, etc.
- Mathematical Functions: ROUND, CEILING, FLOOR, etc.
- Practice: Manipulate names, calculate tenures, format dates.

7. Window (Analytic) Functions

- Concept of Window Functions.
- ROW_NUMBER(), RANK(), DENSE_RANK(), NTILE().
- Aggregate windows: SUM(), AVG() OVER(PARTITION BY ...).
- Practice: Running totals, rankings within groups, percentiles.

8. Views, Indexes & Transactions

- Creating and using Views.
- Understanding & creating Indexes.
- Basics of Transactions (COMMIT, ROLLBACK, SAVEPOINT).

- Practice: Build summary views, measure query performance, simulate transaction scenarios.

9. Data Modeling and Normalization

- 1NF, 2NF, 3NF, BCNF—what, why, and how.
- Creating normalized schemas.
- Practice: Normalize a denormalized table.

10. Stored Procedures, Functions, and Triggers

- Writing and invoking Stored Procedures.
- User-Defined Functions (scalar and table-valued).
- Triggers for automating business logic.
- Practice: Write procedures for HR processes, triggers for audit logging.

11. Advanced Topics

- CTEs (Common Table Expressions) and Recursive CTEs.
- Error Handling & TRY-CATCH/EXCEPTION blocks.
- Performance Tuning: Explain plans, query optimization tips.
- Security: User roles, privileges, GRANT/REVOKE basics.

12. Real-World Projects

- Design and implement a mini HR database with realistic requirements.
- Write complex reports, KPIs, and dashboards using SQL queries.
- Data Migration & ETL basics.

Task Completion and Review

- After each section, submit your scripts and results for review.
- Weekly feedback sessions to clarify doubts and suggest improvements.

Focus on understanding concepts deeply and practicing frequently. If you feel stuck, reach out anytime remember, real growth comes from asking questions and trying out different solutions. Good luck on your journey to becoming a SQL expert!